Detecting Building Changes Using Multimodal Siamese Multitask Networks From Very-High-Resolution Satellite Images

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***Abstract*— Two main issues are faced when using very-high-spatial-resolution (VHR) satellite images for building change detection: 1) the boundaries of detected changes are hard to be consistent with the ground truth and 2) detected changes are easily affected by different viewing angles of bitemporal images, leading to noticeable false changes. To deal with these issues, this study develops a new Siamese change detection network [i.e., Siamese multitask change detection network (SMCD-Net)] based on a multitask learning framework to improve building change detection, particularly in the geometric aspect. Boundary information is formulated as an auxiliary task to constrain the learning of high-level semantic features. To enhance the identification of real changes from false changes, we model the directional relationships between buildings and their shadows by fuzzy sets, and incorporate the relationship information into SMCD-Net, leading to a network variant, labeled as SMCD-Net-m. Experiments were conducted on three datasets: a publicly available dataset, a Chinese GaoFen-2 dataset, and a French Pleiades dataset. We compared our methods with seven other methods, i.e., object-based Siamese network, ChangeStar, ChangeFormer, BIT, STANet, FC-Siam-diff, and Siam-NestedUNet. Results showed that the proposed SMCD-Net obtained the best detection results, achieving the lowest global total errors on all datasets. By incorporating directional information, SMCD-Net-m evidently improved detection accuracy, particularly when using bitemporal images with a large viewing angle difference. The improvement was positively correlated with the accuracy of building shadows extracted from VHR images.**

***Index Terms*— Building change detection, directional relationship modeling, multitask learning, Siamese multitask change detection network (SMCD-Net), Siamese neural network (SNN), very-high-resolution satellite images.**

I. Introduction

MOST of the world's people live in urban areas, where buildings are the primary land cover type used for various purposes, including residence, work, and entertainment. Nowadays, the world is undergoing rapid urbanization, particularly in less developed countries and regions, leading to a fast land change of buildings [1], [2], [3], [4]. Urban sustainable development has been regarded as one of the seventeen goals of the United Nations [5]. To achieve that, updated spatial information on buildings and their changes is essential to policymakers. So far, an increasing number of remote sensing images have been accessible to the public. Among many, it is of particular interest for building change detection using remote sensing images with a very-high-spatial-resolution (VHR), i.e., finer than 1 m, to identify detailed changes [3], [6], [7], [8], [9], [10].

Many methods have been investigated to detect land changes from remote sensing images. A straightforward method is directly comparing the band values of pixels between images acquired at different dates [11]. This study focuses on change detection from bitemporal images. For multispectral images, a transformation like principal component analysis is sometimes used before band comparison [12]. Furthermore, methods based upon change vector analysis (CVA) have also been used in many applications, e.g., detection of land cover changes [13] and forest changes [14]. These pixel-based methods are easy to conduct and commonly applied to low- and medium-resolution remote sensing images. However, they are sensitive to high-frequency image signals, resulting in apparent noise when using VHR images. Moreover, these methods insufficiently use the rich spatial information provided by VHR images.

Object-based image analysis (OBIA) methods have long been preferred to process VHR images [15]. These methods merge a group of homogenous and adjacent pixels to form an image object, corresponding to an image segmentation operation. The image objects as the basic analysis units for further image analysis, e.g., identifying changed objects [16], [17], [18], [19]. Upon image objects, a popular change detection method is first to extract many handcrafted features regarding the object's spectral, textural, and spatial attributes and then to distinguish changed objects from unchanged objects via machine learning algorithms. A comprehensive overview of studies on object-based change detection was given by Hussain et al. [20]. Object-based methods eliminate the so-called "pepper and salt" image noise compared to pixel-based methods and can obtain building change results with more regular boundaries. For complex urban environments, however, it is hard to obtain ideal image segmentation results, in which oversegmented and undersegmented objects may increase the uncertainty of detected changes [21], [22].

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Over the past ten years, deep neural networks have rapidly developed and achieved overwhelming performance than conventional machine learning algorithms in many remote sensing applications [23], [24], [25], [26], [27]. They are tagged with a superb ability to automatically extract high-level image features meaningful for information extraction. Zhang et al. [28] used deep belief networks to extract high-level image features for bitemporal images and then determine land cover changes by computing the feature's cosine angle distance. Alternatively, some studies used machine learning classifiers to determine changes based on extracted features [29]. Motivated by fully convolutional networks (FCNs) that produce an end-to-end semantic segmentation, Zagoruyko and Komodakis [30] and Daudt et al. [31] compared different FCN architectures, i.e., single-branch networks and twin-branch networks [namely, Siamese neural networks (SNNs)], for change detection from images. Their results showed that single-branch networks that concatenate bitemporal images along the channel dimension were inadequate to extract image difference features because features extracted from early layers are apt to be affected by the ill-posed coregistration of pixels from bitemporal images at the same location. In other words, the features extracted from early layers of FCNs may insufficiently characterize the corresponding changes due to a relatively small field of receptive perception for early layers. By contrast, SNNs can avoid this problem and are widely used for remote sensing change detection [32], [33], [34], [35], [36]. Nevertheless, the boundaries of detected building changes produced by existing deep neural networks tend to have a low geometric agreement with real changes. Recently, combining SNN with OBIA improves the geometric accuracy of detected changes [19], [37], [38]. Conventional object-based SNNs, however, are hard to achieve end-to-end detection as they require preliminary image segmentation. Moreover, the detected results by an object-based SNN can be affected by cumulative errors caused by imperfect image segmentation. To seamlessly combine OBIA with deep learning, Zheng et al. [19] proposed a unified deep object-based change detection framework (i.e., ChangeOS) for building damage assessment, in which a building localization module was used to derive building objects. This is an interesting solution to end-to-end detection. Some merits of standard OBIAs (e.g., multiscale analysis for objects with various shapes and sizes), however, are inadequately exploited under this unified framework.

Furthermore, bitemporal VHR images are commonly acquired with inevitable misregistration errors due to different viewing angles. For example, the same building may exhibit different shapes on the VHR images acquired with different viewing angles and has shadows with various conditions. This problem results in noticeable false changes when conducting pixel-by-pixel and object-by-object comparisons directly. Such a problem is more evident for high-rise buildings in complex urban environments and challenges building change detection. Chen and Shi [39] incorporated a pyramid spatial–temporal attention module to SNN for improving the identification of small changes and reducing the influence of illumination variations and misregistration errors on results. However, the effectiveness of this method using bitemporal VHR images with large viewing differences is unexploited. Zhang et al. [40] and de Gélis et al. [41] proposed to use digital surface model (DSM) data together with bitemporal aerial images to improve building change detection. DSM data, however, are not always available for many change detection applications.

This study aims to detect building changes from bitemporal VHR remote sensing images, with a particular interest in improving detection results in the geometric aspect and reducing false changes caused by the difference in viewing angles. We proposed a new SNN based on multitask learning [i.e., Siamese multitask change detection network (SMCD-Net)], in which the boundary information of the changed region of interest is extracted and formulated into an auxiliary task to improve the feature learning of image difference. Previous studies have proved that multitask learning improves the model's generalizability by synthesizing the domain-specific information of related tasks [42] and, thus, improves the geometric accuracy of semantic segmentation results when considering geometric-related tasks [43], [44]. This motivates us to design the proposed SNN in the context of multitask learning. Noticeably, many studies formulated semantic change detection problems into a multitask learning task, in which two key tasks (i.e., semantic segmentation and binary change detection) are learned in a joint representation [45], [46], [47]. By contrast, in this study, building change detection is treated as binary change detection, in which boundary information is jointly learned with the extraction of high-level semantic features to improve the geometric accuracy of detected changes. Moreover, to reduce the false building changes caused by different viewing angles, we assumed that these false changes belong to systematic errors and exhibit directional characteristics, referred to as directional errors. This study quantifies the directional relationship between a building object and its corresponding shadow using fuzzy sets, forming a fuzzy landscape map [48]. The fuzzy landscape maps of bitemporal VHR images are then considered as multimodal data to alleviate directional errors and incorporated into SMCD-Net, leading to a network variant, labeled as SMCD-Net-m. The novelty of this article is given as follows.

1) It proposes a SMCD-Net to improve the accuracy of building change detection, particularly in the geometric agreement between the detected change and the actual change. The proposed model automatically regularizes boundary information during model learning and achieves end-to-end change detection.
2) It incorporates multimodal data that are automatically derived by modeling the directional relationships between building objects and their shadows by fuzzy sets to reduce the false building changes caused by directional errors. A lightweight network is constructed to determine the extent to which the detected building changes belong to false changes.

II. Study Areas and Datasets

We tested the proposed methods using three VHR remote sensing datasets, i.e., a public dataset in Guangzhou (GZ)

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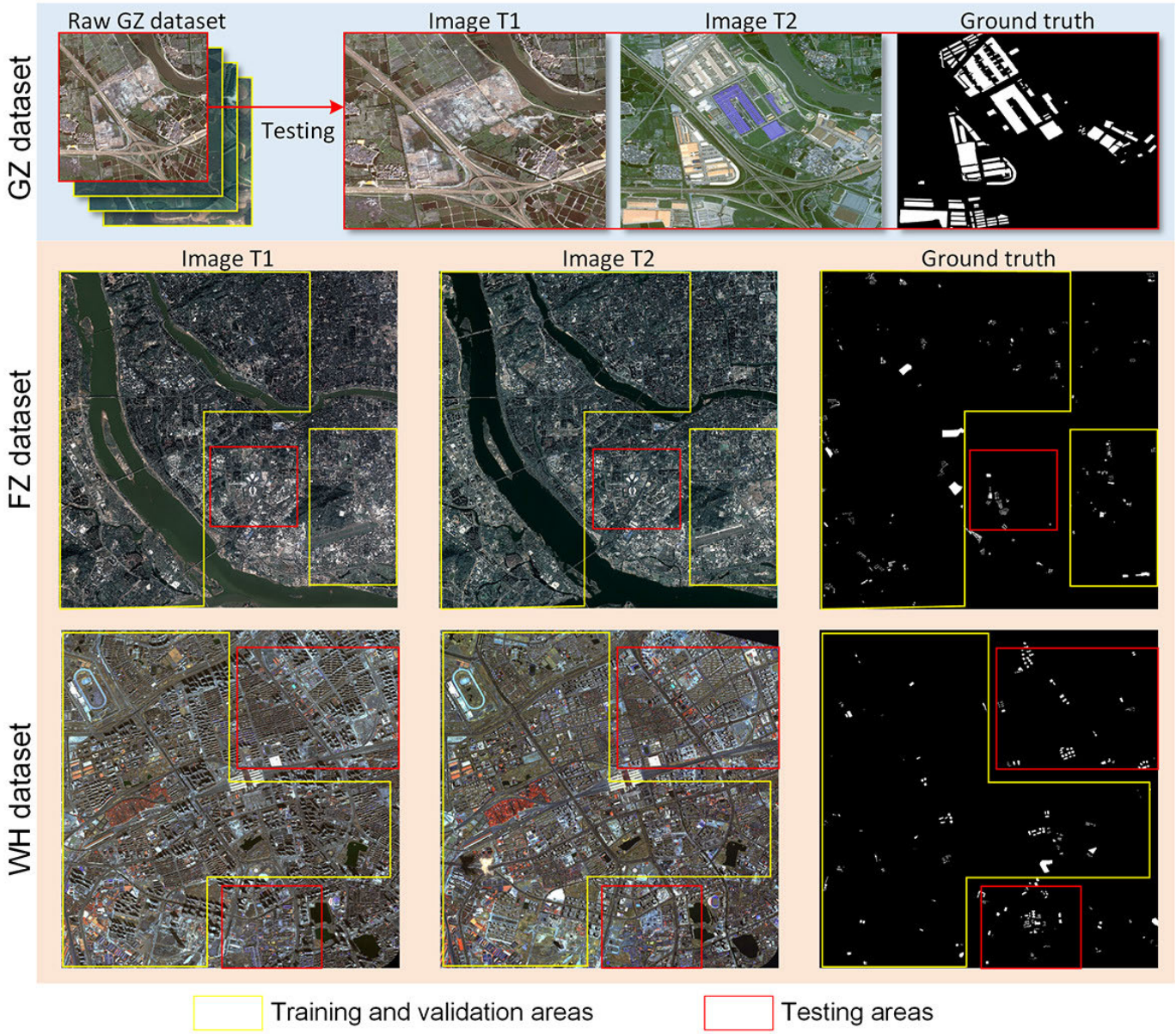


Fig. 1. Overview of bitemporal VHR images on the (public) GZ, FZ, and WH datasets.

TABLE I

DETAILED INFORMATION OF THE GZ, FZ, AND WH STUDY DATASETS. RES., SOLAR AZI., AND SATELLITE AZI. REFER TO SPATIAL RESOLUTION, AND SOLAR AND SATELLITE AZIMUTH ANGLES, RESPECTIVELY

Datasets	Phase	Sensors	Res.(m)	Dates	Solar azi.(°)	Satellite azi.(°)	Sizes(pixels)
GZ	T1	Google Earth	0.55	2006-2019	Unknown	Unknown	From 1006×1168
	T2			2006-2019	Unknown	Unknown	to 4936×5224
FZ	T1	GaoFen-2	0.8	2016.12.07	165.833	95.864	32768×32768
	T2			2020.02.18	152.569	89.208	
WH	T1	Pleiades-1B	0.5	2018.03.12	145.638	97.380	12032×12425
	T2			2019.07.29	125.491	161.151	

(i.e., CD_Data_GZ [49]), a pair of Chinese GaoFen-2 (GF-2) images in FZ, and a pair of French Pleiades images in Wuhan (WH), China, labeled as GZ, FZ, and WH datasets, respectively (see Fig. 1). Table I displays detailed information on these three datasets. GZ, FZ, and WH are the capital cities of Guangdong, Fujian, and Hubei provinces in China, respectively. They have been undergoing rapid urbanization for the past decade, leading to various building changes. The GZ dataset is a popular benchmark for change detection from VHR images. It contains 20 pairs of bitemporal images collected from the Google Earth platform with a spatial resolution of 0.55 m and an image size ranging from 1006 × 1168 to 4936 × 5224 pixels. The FZ and WH datasets, corresponding to two real change detection cases, were created for this research. The FZ dataset contains two GF-2 images, acquired on December 7, 2016, and February 18, 2020. The WH

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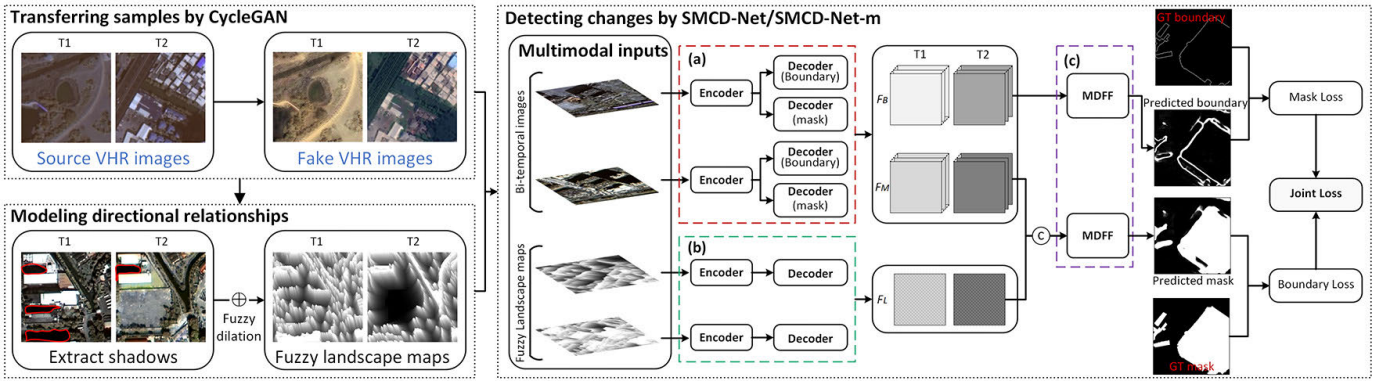


Fig. 2. Workflow of the proposed building change detection. It transfers samples by CycleGAN, models directional relationships by fuzzy landscape maps, and detects building changes by SMCD-Net (or SMCD-Net-m). (a)–(c) Three main components of SMCD-Net, corresponding to a SMFE, a directional feature extractor, and a MDFF module.

dataset contains two Pleiades-1B images, acquired on March 12, 2018, and July 29, 2019. Each GF-2 image includes a panchromatic band of 0.76 m and four multispectral bands of 3.2 m, while each Pleiades image includes a panchromatic band of 0.5 m and four multispectral bands of 2 m. For each VHR image, we fused its panchromatic and multispectral bands using Gram–Schmidt pansharpening and resampled the fused GF-2 images into 0.8 m and Pleiades images into 0.5 m for experiment convenience. To improve change detection, we conducted georeferencing for each pair of bitemporal VHR images on the FZ and WH datasets. Moreover, the raw VHR images in Google Earth are mainly provided by the Worldview and GeoEye satellites launched by the United States, while GF-2 and Pleiades images are typical VHR satellites in China and France. Therefore, these datasets suit to evaluate the performance of our method on VHR images from different sensors.

The ground-truth building changes of the GZ dataset are publicly available, while the ground-truth building changes of the FZ and WH datasets were manually collected. We cropped the VHR images and their corresponding labels on these three datasets into image tiles of 256 × 256 pixels. To train and test our method, we collected training and testing samples separately from each dataset (see Fig. 1). The training samples were further divided into training and validation sets. In total, we obtained 1600, 2550, and 600 raw samples (in image pair) for the GZ, FZ, and WH datasets, respectively. We then performed sample enhancement to these samples, including rotation, mirroring, and scaling operations. Finally, we obtained 3310, 6000, and 2000 samples for the GZ, FZ, and WH datasets. The ratio of training, validation, and testing sets of each dataset was specified as (0.7:0.1:0.2). The selected testing areas in these three datasets contain various building changes, suiting performance evaluation.

III. Methodology

The workflow of the proposed building change detection using SMCD-Net is illustrated in Fig. 2. We constructed a SNN based on multitask learning, in which the building's boundary information was jointly learned together with the information of the building mask. By doing so, the geometric accuracy of the detected building changes is improved. Second, for each VHR image, we modeled the directional relationships between a building and the corresponding shadow by fuzzy sets, leading to a fuzzy landscape map. The fuzzy landscape maps of bitemporal images were then used as additional modal data, helping to reduce the systematic errors [i.e., false positive (FP)] of detected changes caused by the shadow effect. Multimodal difference features are then fused at multilevels for change detection. Finally, we used a GAN to generate target-like samples from sample-rich datasets to construct a relatively sufficient training dataset for the proposed model.

A. Siamese Multitask Feature Extractor for Building Change Detection

The structure of the proposed Siamese multitask feature extractor (SMFE) is shown in Fig. 3(a). SMFE focuses on learning high-level image features regarding the mask and boundary information of potential changes for bitemporal images. The two types of features are learned jointly based on a multitask framework. SMFE comprises two identical multitask feature extractors (MFEs) with shared weights that learn image features for images at T_1 and T_2. The backbone network of MFE adopts a typical encoder–decoder architecture, consisting of an encoder and two decoders [see Fig. 3(a)]. In complex urban environments, buildings are usually characterized by various roof materials, shapes, and sizes, posing evident large intraclass dissimilarity and small interclass dissimilarity (compared with other land cover elements). Therefore, the identification of building changes demands exploiting multiscale contextual information, particularly in areas more likely to be changed. In this study, we improved the backbone network based on a regular UNet from two aspects. We replaced convolutional layers in the feature extractor with hybrid dilated convolutions (HDCs) for expanding the receptive field of the SMCD-Net without increasing the number of model parameters. An atrous spatial pyramid pooling (ASPP) is then added to the last downsampling operation of the encoder. The ASPP uses different dilated rates in a convolution layer to obtain and connect multilevel feature maps. Moreover, a convolutional block attention module (CBAM) [50] is added at the last layer of the encoder. By doing so, the receptive field of SMCD-Net

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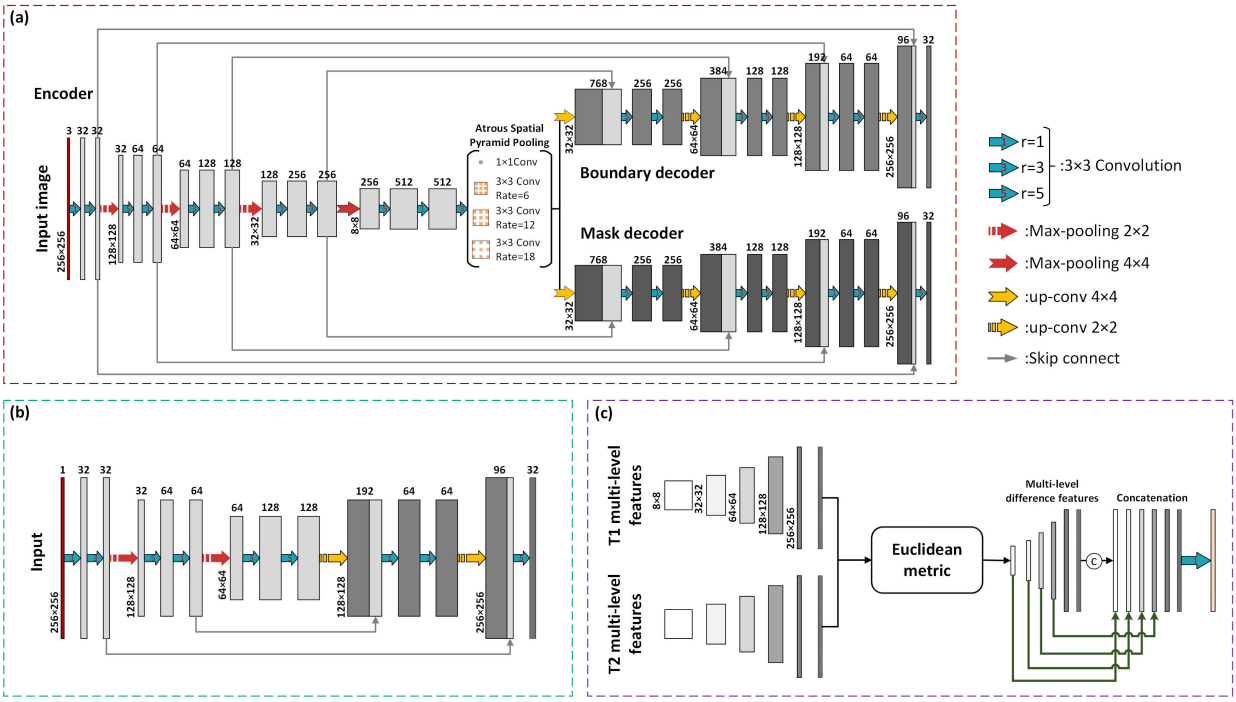


Fig. 3. Details of (a) MFE, (b) directional feature extractor, and (c) MDFF module.

can be expanded to enhance the exploitation of multiscale contextual information for change detection. Incorporating attention modules further improves the separability between changed and nonchanged buildings.

More specifically, the encoder of a MFE includes five downsampling modules, each consisting of a convolution layer (kernel $= 3 \times 3$, stride $= 1$, and dilated $= [1, 3, 5]$), a max-pooling layer, and an ReLU activation function. By downsampling, the number of feature channels continually increases by convolution operations, leading to 32, 64, 128, 256, and 512. For a VHR image at T_1 (resp. T_2), the MFE encoder extracts a set of image features $F_l^{T_1}$ (resp. $F_l^{T_2}$) with $l = 1, 5$ by the downsampling modules.

Each MFE has two decoders with the same structure, corresponding to mask and boundary predictions. The encoder-derived features $F_l^{T_1}$ (resp. $F_l^{T_2}$) are piped into upsampling operations, generating multilevel mask features $F_{M_l}^{T_1}$ (resp. $F_{M_l}^{T_2}$) and boundary features $F_{B_l}^{T_1}$ (resp. $F_{M_l}^{T_2}$). To fully use shallow features of VHR images, e.g., texture and geometry, the decoders adopt "skipping connection" to combine hierarchical features of bitemporal images to improve feature extraction. Here, the mask and boundary predictions are seen as two subtasks of SMFE. The incorporation of boundary information constrains the changed areas detected by SMFE with regularized boundaries, improving the geometric accuracy of these areas.

B. Extraction of Directional Difference Features

As illustrated in Fig. 4, we found that false building changes caused by images with different off-nadir angles (or solar azimuth ϕ) have a clear directional relation with respect to building shadows. To alleviate these false changes, an immediate solution is defining a function $\mu = f(x_{CD}, s_{CD}, \beta_1, \beta_2)$ to model the detected building changes x_{CD} with shadow changes s_{CD} given image reviewing angles β_1 and β_2. However, such a function is hard to explicitly construct. In this article, we proposed to decompose the above function into a combination $\mu = g(\mu_1, \mu_2)$ of modeling the directional relationship μ_1 (resp. μ_2) between a building pixel x_b^1 (resp. x_b^2) and the corresponding shadow pixel x_s^1 (resp. x_s^2) on image T_1 (resp. T_2) with angle β_1 (resp. β_2). Motivated by Li et al. [48], we quantified the directional relationships μ_1 and μ_2 by fuzzy sets into fuzzy landscape maps. Moreover, we designed a lightweight network to extract high-level directional features based on fuzzy landscape maps to determine the extent to which the detected building changes belong to false changes [see Fig. 3(b)].

1) Characterizing Directional Relationships by Fuzzy Landscape: We first applied a method based on Otsu thresholding to automatically extract building shadows from a VHR multispectral image [51], [52]. More specifically, we converted the image's near-infrared-red-green false color space to a hue–saturation–intensity (HSI) color space, upon which a $(\text{Saturation} - \text{Intensity})/(\text{Saturation} + \text{Intensity})$ index is formulated for Otsu thresholding. To eliminate the vegetation effect, we removed vegetation areas based on the normalized difference vegetation index (NDVI) using Otsu thresholding.

Given a shadow object R_s and a direction angle β, where β is inferred from solar azimuth angle ϕ, i.e., $\beta = \phi - (\pi/2)$, we defined a membership function $L_\beta(x, R_s)$ to quantify the degree of an image pixel $x \in I$ satisfying the directional relationship to R_s. We measured the membership value by taking angle $\theta_\beta(x, b)$ and distance $d(x, b)$ terms into account, where $b \in R_s$. $\theta_\beta(x, b)$ specifies the angle between the vector from x to b and the unit vector along the direction angle β

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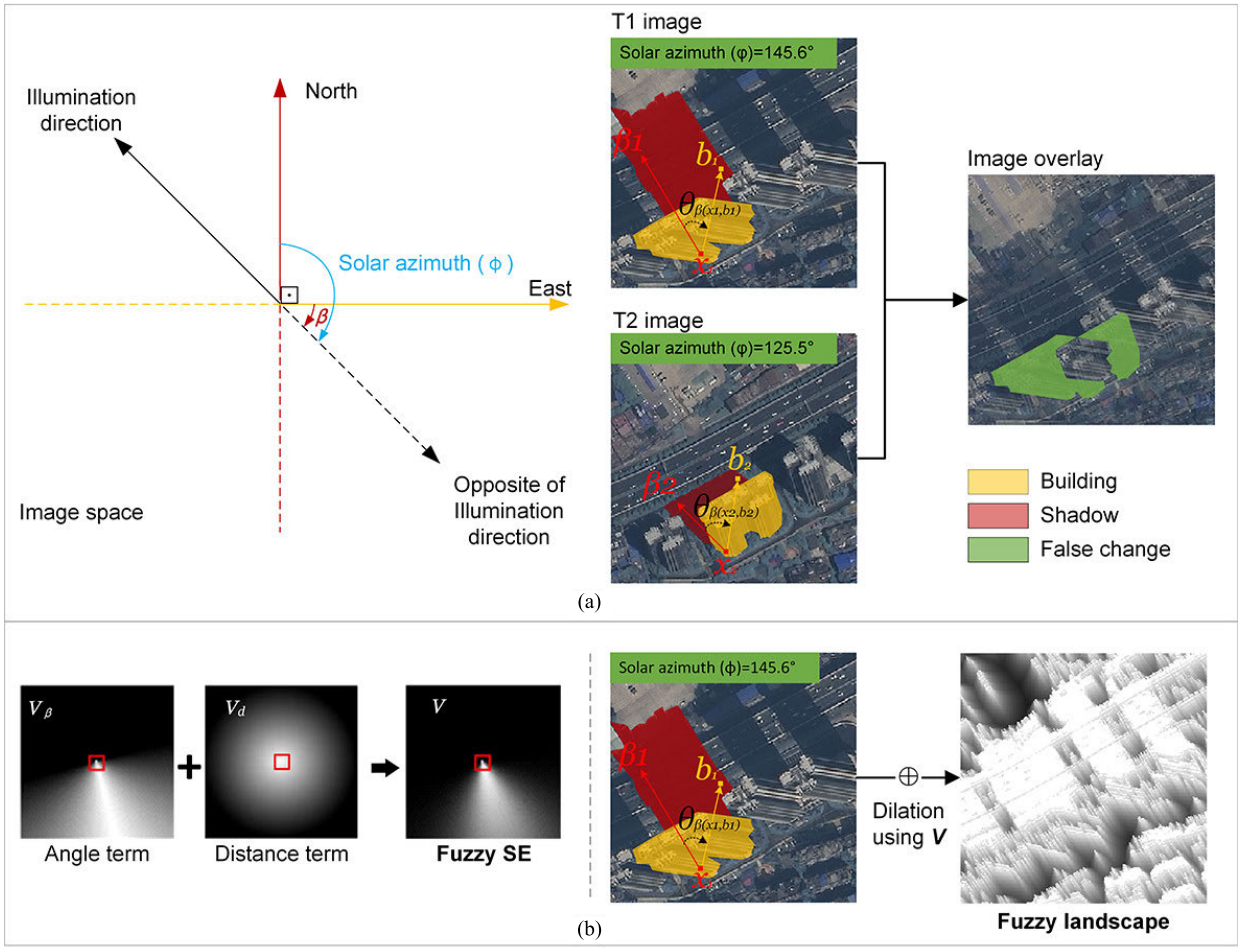


Fig. 4. Modeling directional relationships between buildings and their shadows. (a) Viewing angle β of an image and the false changes caused by bitemporal images with different viewing angles. (b) Creation of a fuzzy SE and a fuzzy landscape based on building shadows by a dilation operation $\oplus$.

of the horizontal axis, while $d(x, b)$ specifies the Euclidean distance between x and b. Following Li et al. [48], we used fuzzy morphological dilation to derive the fuzzy landscape. Moreover, we used the boundary pixels δR_s of R_s for the dilation operation to improve the computation efficiency. The fuzzy landscape $L_\beta(x, R_s)$ is then derived from

$$L_\beta(x, R_s) = (\delta R_s \oplus v)(x) \bigcap {R_s}^c \tag{1}$$

where ${R_s}^c$ is the complement of R_s and $\oplus$ indicates a dilation operator using a fuzzy structuring element (SE) v. v is formulated by an angle-related SE v_β and a distance-related SE v_d

$$v = v_\beta(x) \times v_d(x) \tag{2}$$

$$v_\beta(x) = \max\left(0, 1 - \frac{2}{\pi}\theta_\beta(x, b)\right) \tag{3}$$

$$v_d(x) = \max\left(0, 1 - \frac{2 \times d(x, b)}{\sqrt{2} \times k}\right). \tag{4}$$

2) Directional Feature Extraction: We constructed a directional feature extractor to obtain high-level directional features associated with building shadow based on the fuzzy landscape of a VHR image. Considering the fuzzy landscape is only a single-channel feature map with a strong characterization for each pixel, we focused on extracting high-level contextual features for the fuzzy landscape map. Here, we designed an encoder–decoder feature extractor with shallow convolutional layers [see Fig. 3(b)]. The encoder contains two downsampling modules, in which each module performs two 3×3 convolutions and a 2×2 max pooling. The decoder also has two upsampling modules performing four 3×3 convolutions, two 2×2 up-conv operations, and a 1×1 convolution.

Using the extractor, we obtained two directional feature maps, i.e., $F_L^{T_1}$ and $F_L^{T_2}$, for bitemporal VHR images T_1 and T_2. We then treated these two feature maps as additional information for change detection.

3) Multilevel Difference Feature Fusion for Multimodal Data: For mask features $F_{M_l}^{T_1}$ and $F_{M_l}^{T_2}$ at the lth feature extraction level, we obtained a mask difference feature $\mathrm{DF}_{\mathrm{Mask}_l}$ by computing their Euclidean distance $\mathrm{DF}_{\mathrm{Mask}_l} = (|F_{\mathrm{Mask}_l}^{T_1} - F_{\mathrm{Mask}_l}^{T_2}|^2)^{1/2}$. We designed a multilevel difference feature fusion (MDFF) module to fuse difference features at multilevels [see Fig. 3(c)]. The MDFF module consists of a set of upsampling operations to reshape multilevel difference map $\mathrm{DF}_{\mathrm{Mask}_l}$ with the same size of the finest feature map (i.e., $\mathrm{DF}_{\mathrm{Mask}_5}$, followed by a concatenation operation and a 1×1 convolution). It then outputs a mask difference map $\mathrm{DF}_{\mathrm{Mask}}$. Similarly, we obtained a boundary difference map $\mathrm{DF}_{\mathrm{Boundary}}$.

Furthermore, the MDFF module can be adjusted to incorporate the directional features $F_L^{T_1}$ and $F_L^{T_2}$. Here, the Euclidean

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difference between $F_L^{T_1}$ and $F_L^{T_2}$ is computed and concatenated at the end of $\mathrm{DF}_{\mathrm{Mask}_l}$, as shown in Fig. 2(c).

Based on $\mathrm{DF}_{\mathrm{Mask}}$ and $\mathrm{DF}_{\mathrm{Boundary}}$, mask and boundary change maps can be derived using a softmax activation function and a threshold equal to 0.5. The use of MDFF retains comparing feature differences at earlier levels, facilitating to identifying changes at multiple levels and then improving final change detection performance.

4) Joint Loss Function: We first computed two separate losses for the change mask and boundary tasks. A joint loss is then derived by combining the two losses. We used a focal loss [53] for the mask task. It is an improved version of the cross-entropy loss function. It can guide the model to focus more on changed samples by lowering the weight of unchanged samples, improving the unbalance between changed and unchanged samples [54]. The mask loss L_{mask} is defined as

$$L_{\mathrm{mask}} = -\alpha(1-\rho_t)^{\lambda}\log(\rho_t) \tag{5}$$

where α controls the shared weight of positive and negative samples to the total loss. λ refers to a modulating factor to make the model focus on examples that are hard to classify by reducing the weights of the examples that are easy to classify. In our study, α is set to 0.25, and λ is set to 2.

We used a negative log-likelihood (NLL) [55] loss for boundary task

$$L_{\mathrm{boundary}} = \sum_{x\in\omega}\log p_{\mathrm{boundary}}\left(x; L_{\mathrm{boundary}}(x)\right) \tag{6}$$

where L_{boundary} represents the pixel classification error and x refers to a pixel position in the image space ω. $\log p_{\mathrm{boundary}}(x; L_{\mathrm{boundary}}(x)$ represents the prediction probability of true label L_{boundary}.

Based on the mask and boundary losses L_{mask} and L_{boundary}, we obtained a joint loss $L_{\mathrm{Siam_mul_task}}$

$$L_{\mathrm{Siam_mul_task}} = \theta L_{\mathrm{mask}} + (1-\theta)L_{\mathrm{boundary}} \tag{7}$$

where θ is a weight factor balancing the importance of the above two losses. Here, θ is determined by grid searching from 0 to 1 with a step of 0.1 [56].

C. Sample Transferring Based on Generating Adversarial Networks

To enrich the training dataset, we used a cycle generative adversarial network (CycleGAN) to generate target samples from sample-rich datasets [57]. Compared with other generative adversarial networks, CycleGAN does not require pairwise matching of training data. In addition, it can interconvert source-domain data with target-domain data so that it can enrich two datasets by only training one model. The CycleGAN consists of two symmetric GANs, each containing a generator and a discriminator (see Fig. 5). The discriminator of CycleGAN calculates the degree of difference between the data produced by the generator and the target domain data. Based on the degree of difference, the generator gradually adjusts a random noise distribution to a true distribution of the target domain data.

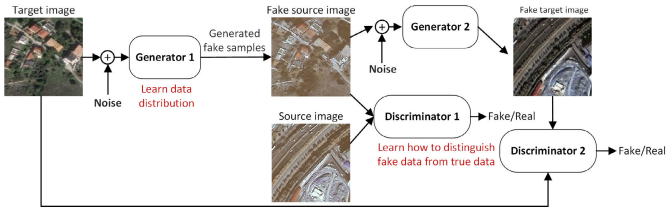


Fig. 5. Demonstration of using CycleGAN for sample transfer.

We used the target domain samples and available samples (source samples) from other sensors to constrain the CycleGAN. The source samples are then transformed into a pseudotarget domain sample using the fully trained CycleGAN. Finally, we utilized target domain and pseudotarget domain samples to constitute a samples' library.

D. Accuracy Assessment and Performance Evaluation

To differentiate the proposed model using and without using the directional relationship information derived from fuzzy landscape maps, we labeled SMCD-Net-m for the model using directional information and SMCD-Net otherwise. We evaluated the accuracy of the detected building changes by our method using both attribute and geometric measures. We used precision (P), recall (R), and F1-score (F1) measures to evaluate attribute accuracy

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{8}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{9}$$

$$\mathrm{F1} = 2\times\frac{P\times R}{P+R} \tag{10}$$

where TP, FP, and FN indicate true positive, false positive, and false negative, respectively. Besides, we used global overclassification (GOC) error, global underclassification (GUC) error, and global total classification (GTC) error indices to evaluate geometric accuracy [48]. More specifically, let M_i be a detected change object, O_{ij} with $j = 1, 2, \ldots, r$, be a set of reference objects, where $r = 1$ and $r > 1$ refer to one-to-one and one-to-many comparison cases, respectively [48]. Let area(M_i) and area(O_{ij}) be the area of M_i and O_{ij}, and area$(M_i \cap O_{ij})$ be the overlapping area of the pair of objects (M_i, O_{ij}). The overclassification $\mathrm{OC}(M_i)$ and underclassification $\mathrm{UC}(M_i)$ errors are defined as

$$\mathrm{OC}(M_i) = \sum_{j=1}^{r}\left(w\cdot\left(1-\frac{\mathrm{area}\left(M_i\cap O_{ij}\right)}{\mathrm{area}\left(O_{ij}\right)}\right)\right) \tag{11}$$

$$w = \frac{\mathrm{area}\left(O_{ij}\right)}{\sum_{j=1}^{r}\mathrm{area}\left(O_{ij}\right)} \tag{12}$$

$$\mathrm{UC}(M_i) = 1-\frac{\sum_{j=1}^{r}\mathrm{area}\left(M_i\cap O_{ij}\right)}{\mathrm{area}(M_i)}. \tag{13}$$

The total classification (TC) error $\mathrm{TC}(M_i)$ is derived by combining $\mathrm{OC}(M_i)$ with $\mathrm{UC}(M_i)$

$$\mathrm{TC}(M_i) = \sqrt{\frac{\mathrm{OC}(M_i)^2+\mathrm{UC}(M_i)^2}{2}}. \tag{14}$$

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Based on $\mathrm{OC}(M_i)$, $\mathrm{UC}(M_i)$, and $\mathrm{TC}(M_i)$, three global error indices, i.e., GOC, GUC, and GTC, are derived

$$\mathrm{GOC} = \sum_{i=1}^{m} \left(\mathrm{OC}(M_i) \cdot \frac{\mathrm{area}(M_i)}{\sum_i^m \mathrm{area}(M_i)} \right) \tag{15}$$

$$\mathrm{GUC} = \sum_{i=1}^{m} \left(\mathrm{UC}(M_i) \cdot \frac{\mathrm{area}(M_i)}{\sum_i^m \mathrm{area}(M_i)} \right) \tag{16}$$

$$\mathrm{GTC} = \sum_{i=1}^{m} \left(\mathrm{TC}(M_i) \cdot \frac{\mathrm{area}(M_i)}{\sum_i^m \mathrm{area}(M_i)} \right) \tag{17}$$

where m is the number of detected building change objects.

To evaluate the performance of our method, we compared it with seven recent methods, including four CNN-based, an object-based SNN, i.e., Obj-SiamNet [37], and three pixel-level change detection methods, i.e., FC-Siam-diff [31], Siam-NestedUNet [32], [58], STANet [39], and ChangeStar [9], and two transformer-based methods, namely, ChangeFormer [59] and BIT [60].

Noticeably, Obj-SiamNet applies OBIA to SNN, leading to an evident improvement in geometric accuracy. The architecture of the Obj-SiamNet is given in Fig. A1 in the Appendix. It consists of four main steps: multiscale image segmentation, Siamese feature extraction, difference feature calculation, and multiscale change information fusion. Extracted deep features of bitemporal images were first used to derive change information at the pixel level, which was further converted into object-level based on image segments. Object-based change information at multiple scales is then fused to obtain final changes by fuzzy sets. For a fair comparison, the ChangeStar based on bitemporal supervised learning was implemented for building change detection.

E. Implementation Details

We applied our method to three building change datasets, i.e., GZ, FZ, and WH datasets, and trained it for 100 epochs using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 16. Experiments were performed on an NVIDIA GeForce RTX 2080Ti GPU. It is worth mentioning that we adopted a multiscale sliding window prediction to eliminate the edge effect, where the mean value of the predicted multiscale images was used. Specifically, the sizes of predicted multiscale images are 256 × 256, 512 × 512, and 768 × 768 pixels, respectively, and the corresponding sliding window step size is 256, and the repetition rate corresponds to 0, 1/2, and 1/3. Here, we only considered the central part (of 256 × 256 pixels) of multiscale images to calculate their mean values. The same implementation settings were used for change detection by other existing methods.

IV. Results

A. Ablation Experiments

To validate the effectiveness of different modules (i.e., SMFE, HDC-CBAM, and MDFF) for building change detection, we conducted ablation experiments on the GZ dataset. We sequentially removed these modules from SMCD-Net, leading to three model variants. We used a focal loss function, and the same hyperparameters and samples described in Section III-E of the methodology to train all these model variants for change detection. Table II lists the accuracy of detection results from the ablation experiments. The accuracy of detection decreases evidently without using these modules, in which the SMFE feature extractor significantly influences change detection performance. Fig. 6 shows the results obtained from the ablation experiments. Comparing Fig. 6(a) and (b), the SMFE module helps to detect buildings with regularized shapes, especially for complex buildings [see boxes A and B in Fig. 6(a) and (b)]. The use of the HDC-CBAM module improves the detection of small changed objects and reduces detection errors [see Fig. 6(b) versus (c)], while the MDFF module further improves detection accuracy [see Fig. 6(c) versus (d)]. It is obvious that the poorest detection result was obtained using the model without these three modules [see Fig. 6(d)].

TABLE II

Accuracy Evaluation of Ablation Experiments. Contribution, Labeled as C-F1 and Defined Based on F1-Score, Calculates the Decrement Ratio by Removing One Module to the Total Decrement When Removing All Modules. H-C Refers to HDC and CBAM

	SMFE	H-C	MDFF	GTC	P	F1	C-F1 ↓
a (ours)	✓	✓	✓	**0.079**	**0.954**	**0.917**	
b	×	✓	✓	0.135	0.893	0.835	**61.65%**
c	×	×	✓	0.137	0.883	0.798	27.82%
d	×	×	×	0.168	0.838	0.784	10.53%

B. Building Change Detection by SMCD-Net on the Public GZ Dataset

We first applied SMCD-Net (without using directional information) to detect building changes on the public GZ dataset and compared it with seven other methods (i.e., Obj-SiamNet, ChangeStar, ChangeFormer, BIT, STANet, Siam-NestedUNet, and FC-Siam-diff). Images of this dataset were already preprocessed into red-green-blue (RGB) composite without containing metadata information for sun azimuth angles. Hence, we only used the RGB channels for image change detection. We used 3310 pairs of GZ samples, with 70% for training, 10% for validation, and 20% for testing.

Table III lists the accuracy of building change detection in attribute and geometric measures. This table shows that SMCD-Net performed better than all other methods for all accuracy measures except the GUC error. More specifically, it produced results with an F1 of 0.917, which is evidently higher than the second-best results obtained by ChangeStar, i.e., 0.892. The results of SMCD-Net have a GTC error of 0.079, while ChangeStar produced the second lowest GTC error of 0.087. Fig. 7 compares the detection results of SMCD-Net with others on the public GZ dataset. This figure shows that the obtained results from SMCD-Net are highly consistent with the ground-truth changes, producing changed buildings with more regularized boundaries [red box A in Fig. 7(a1)] than that of ChangeStar, ChangeFormer, BIT, STANet, Siam-NestedUNet, and FC-Siam-diff results.

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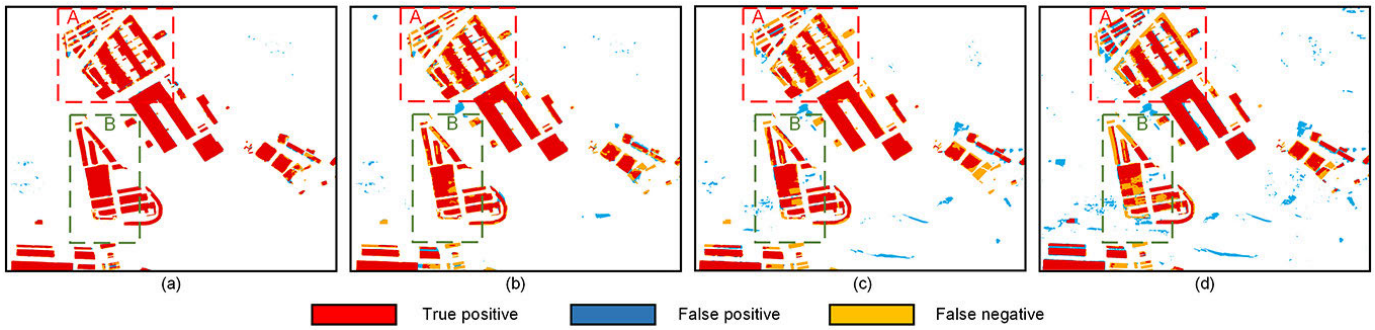


Fig. 6. Comparison of change detection results using different modules in ablation experiments. (a) Results obtained by SMCD-Net. (b)–(d) Results obtained by the three SMCD-Net variants without SMFE, without SMFE and HDC-CBAM, and without SMFE, HDC-CBAM, and MDFF modules, respectively.

TABLE III

EVALUATION OF THE RESULTS OF BUILDING CHANGE DETECTION BY DIFFERENT METHODS ON THE PUBLIC GZ DATASET USING ATTRIBUTE AND GEOMETRIC MEASURES. GOC, GUC, GTC, P, R, AND F1 REFER TO THE GLOBAL OVERCLASSIFICATION, GLOBAL UNDERCLASSIFICATION, GLOBAL TOTAL CLASSIFICATION ERRORS, PRECISION, RECALL, AND F1-SCORE

Methods	GOC	GUC	GTC	P	R	F1
SMCD-Net	**0.023**	0.106	**0.079**	**0.954**	**0.891**	**0.917**
Obj-SiamNet	0.100	0.133	0.134	0.892	0.834	0.862
ChangeStar	0.068	0.079	0.087	0.924	0.863	0.892
ChangeFormer	0.108	0.106	0.121	0.875	0.854	0.864
BIT	0.073	0.211	0.165	0.894	0.761	0.822
STANet	0.087	0.157	0.141	0.881	0.792	0.834
Siam-NestedUNet	0.359	**0.067**	0.265	0.469	0.803	0.592
FC-Siam-diff	0.075	0.192	0.168	0.838	0.736	0.784

The poorest results were obtained by Siamese-NestedUNet, followed by BIT and STANet, for which obvious change areas were undetected or detected with fragmented objects and ragged boundaries. Obj-SiamNet also produced relatively regularized building boundaries, implying that incorporating OBIA with SNN can improve change detection. However, it has difficulties in separating small buildings arranged closely [yellow box B in Fig. 7(b1)]. The same problem was found in Changeformer and Siam-NestedUNet. From the distribution of the TC error of different detection results, we can see that SMCD-Net obtained the best results than other methods in geometric agreement with the ground truth.

C. Building Change Detection on the FZ and WH Datasets

We then applied the proposed SMCD-Net to detect building changes on the FZ and WH datasets, and compared it with seven other methods. The images in these two datasets were in RGB composite. Fig. 8 shows change detection results by different methods on the FZ dataset and the corresponding distribution of TC errors. The changed areas detected by SMCD-Net are generally more consistent with the ground-truth than other methods.

More specifically, SMCD-Net outperforms others in detecting changes in densely arranged buildings [red box A in Fig. 8(a1)–(h1)]. In contrast, results obtained by ChangeStar, ChangeFormer, BIT, STANet, Siam-NestedUNet, and FC-Siam-diff have evident false negatives (i.e., detection omission) [boxes B and C in Fig. 8(c1)–(h1)], while the detected changes by Obj-SiamNet tend to be connected for densely arranged buildings. Table IV lists the accuracy of detection results by different methods. The proposed SMCD-Net achieved the highest F1 of 0.754 and the lowest GTC error of 0.213 than other methods, followed by the second-best result by ChangeStar with F1 of 0.681 and GTC error of 0.215.

TABLE IV

EVALUATION OF THE CHANGES DETECTED BY DIFFERENT METHODS ON THE FZ DATASET USING ATTRIBUTE AND GEOMETRIC MEASURES

Methods	GOC	GUC	GTC	P	R	F1
SMCD-Net	0.119	0.236	**0.213**	0.774	**0.737**	**0.754**
Obj-SiamNet	0.181	0.249	0.247	0.706	0.623	0.662
ChangeStar	0.057	**0.211**	0.215	0.812	0.587	0.681
ChangeFormer	**0.043**	0.346	0.254	**0.842**	0.552	0.665
BIT	0.181	0.364	0.338	0.769	0.550	0.641
STANet	0.170	0.324	0.278	0.812	0.465	0.592
Siam-NestedUNet	0.127	0.530	0.267	0.610	0.394	0.479
FC-Siam-diff	0.062	0.436	0.320	0.836	0.445	0.581

Next, we applied SMCD-Net to detect building changes from a pair of Pleiades images on the WH dataset with two testing areas, i.e., Wuhan A (WH-A) and Wuhan B (WH-B). The changed areas account for a small portion, leading to an insufficient number of samples that could be collected from this dataset. Here, we used CycleGAN to transfer the collected samples from the FZ dataset into the WH domain to increase the number of WH samples. We randomly selected 600 samples for each of the FZ and WH training areas to train the CycleGAN. We set the batch size to 16, the epoch to 150, and the learning rate to 10^{-3}. The CycleGAN transferred 2000 samples from the FZ dataset to the WH domain. These transferred samples were used together with those WH samples to train SMCD-Net on the WH dataset. We evaluated the effectiveness of using CycleGAN to transfer samples across different domains. We compared the results of change detection by SMCD-Net in the WH-A and WH-B areas between using and without using transferred samples. Fig. 9 shows that the detection using transferred samples has better results than those without transferred samples. The same evidence is obtained from the accuracy evaluation results in Table V.

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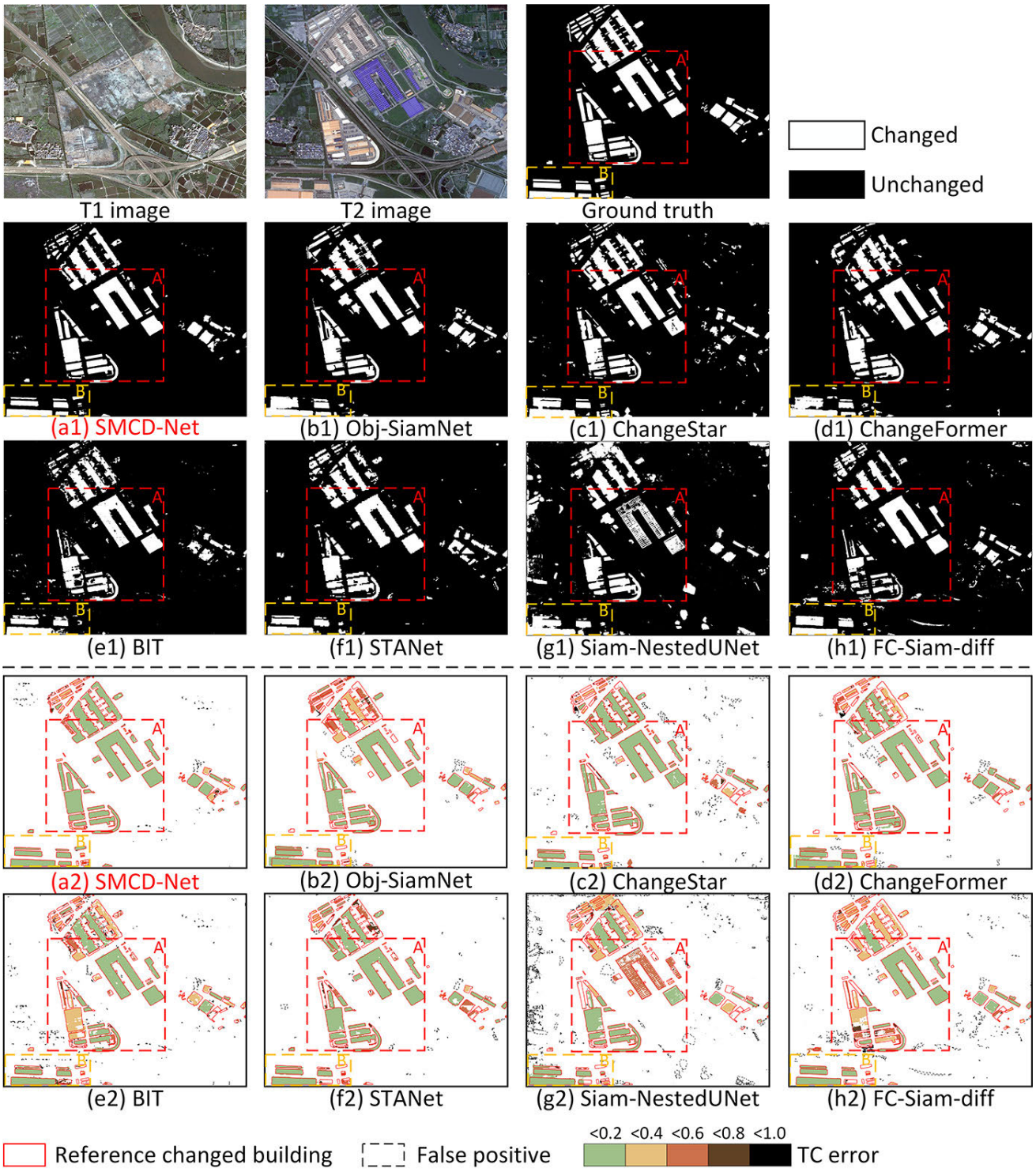


Fig. 7. (Top) Comparison of building change detection results between different methods on the public GZ dataset. (Bottom) Distribution of the TC error of different detection results. (a1) and (a2) SMCD-Net. (b1) and (b2) Obj-SiamNet. (c1) and (c2) ChangeStar. (d1) and (d2) ChangeFormer. (e1) and (e2) BIT. (f1) and (f2) STANet. (g1) and (g2) Siam-NestedUNet. (h1) and (h2) FC-Siam-diff.

We compared the building change detection results of SMCD-Net with the other seven methods in the WH-A and WH-B testing areas (see Figs. 10 and 11). Fig. 10 shows that the proposed SMCD-Net obtained the best detection results in the WH-A area, followed by ChangeStar and Obj-SiamNet. More specifically, SMCD-Net detected building changes with more regularized boundaries and had superiority in detecting small and irregular building changes. Nevertheless, we can see that all methods produced results with noticeable false changes in the WH-B area, especially for high-rise changed buildings, as shown in box D in Fig. 11(a1)–(h1). The false changes might be caused by obvious spatial displacement due

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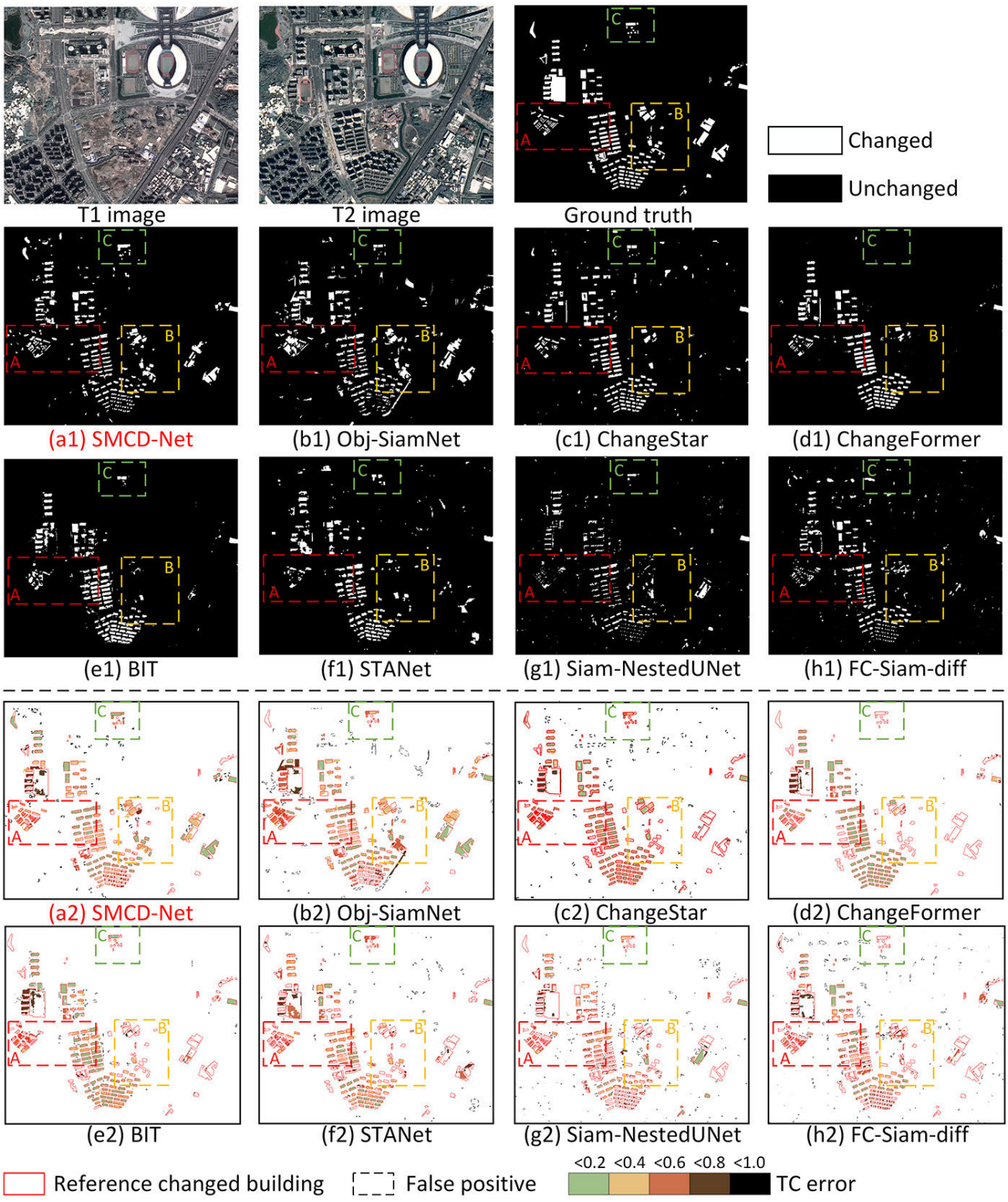


Fig. 8. (Top) Comparison of building change detection results between different methods on the FZ dataset. (Bottom) Distribution of the TC error of different detection results. (a1) and (a2) SMCD-Net. (b1) and (b2) Obj-SiamNet. (c1) and (c2) ChangeStar. (d1) and (d2) ChangeFormer. (e1) and (e2) BIT. (f1) and (f2) STANet. (g1) and (g2) Siam-NestedUNet. (h1) and (h2) FC-Siam-diff.

to different viewing angles between the bitemporal images. Table VI lists the accuracy of detection results by different methods in WH-A and WH-B testing areas. The results obtained by SMCD-Net have the highest F1 and the lowest GTC error than others for both areas.

D. Modeling Directional Relationships as Multimodal Information

To reduce the false changes caused by the difference in viewing angles of bitemporal images, we modeled the directional relationship between buildings and their shadows by the

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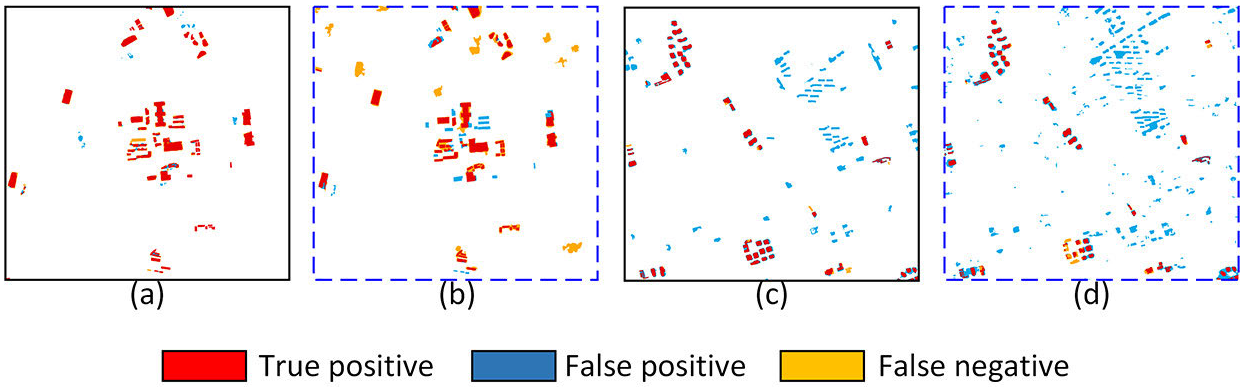


Fig. 9. Comparison of the results of change detection in WH-A and WH-B testing areas by SMCD-Net using and without using transferred samples from the FZ study area. (a) and (c) Change detection in WH-A and WH-B areas with transferred samples. (b) and (d) Change detection in WH-A and WH-B areas without transferred samples.

TABLE V
EVALUATION OF THE RESULTS OF CHANGE DETECTION IN THE WH-A AND WH-B TESTING AREAS BY SMCD-NET USING AND WITHOUT USING TRANSFERRED SAMPLES FROM THE FZ STUDY AREA USING BOTH ATTRIBUTE INDICES AND GEOMETRIC ERRORS. TS REFERS TO TRANSFERRED SAMPLES

Areas	Methods	GOC	GUC	GTC	P	R	F1
WH-A	Without TS	0.178	0.043	0.137	0.529	0.807	0.639
	With TS	**0.057**	**0.139**	**0.113**	**0.894**	**0.823**	**0.857**
WH-B	Without TS	0.265	0.169	0.256	0.441	0.581	0.502
	With TS	**0.149**	**0.018**	**0.109**	**0.514**	**0.901**	**0.650**

fuzzy landscape. The fuzzy landscape map of a VHR image was used as multimodal information for change detection. Fig. 12 shows the fuzzy landscape maps extracted from bitemporal VHR images in both FZ and WH-B testing areas. The value of a fuzzy landscape ranges from 0 to 1, where a larger value indicates a stronger directional relationship, and vice versa.

To differentiate the proposed SMCD-Net trained only using VHR images, we named the method incorporating directional information as SMCD-Net-m. Since the WH-A testing area was dominated by low buildings without noticeable building displacements, here, we chose the Pleiades bitemporal images in the WH-B area with evident building displacements to test the performance of SMCD-Net-m. Moreover, the GF2 images in the FZ testing area were also used to test the applicability of SMCD-Net-m for different sensors. We derived the fuzzy landscape map for each VHR image and incorporated it to train SMCD-Net-m using the same examples and parameters as the above experiments. Fig. 13 compared the detection results between SMCD-Net-m and SMCD-Net. This figure shows that SMCD-Net-m (incorporating directional information) significantly improved detection results by reducing false changes caused by the difference in viewing angles, as can be seen from box A in Fig. 13(b) and (c) and box B in Fig. 13(e) and (f). Table VII compares the accuracy of the detected building changes between SMCD-Net-m and SMCD-Net in the FZ and WH-B testing areas. This table shows that the results obtained by SMCD-Net-m have a higher F1 and a lower GTC error than that of SMCD-Net on both the two areas, implying that SMCD-Net-m further improved SMCD-Net when incorporating directional information.

To test the effectiveness of incorporating directional information by the proposed strategy [i.e., using a lightweight network; see Fig. 2(b)], we compared SMCD-Net-m with a straightforward strategy that directly concatenates the directional information to the input RGB images as additional channels (see Fig. 14). This figure shows that SMCD-Net-m performed much better than the straightforward strategy in reducing false changes, implying the effectiveness of the proposed strategy.

E. Effect of Shadow Extraction on Change Detection Results

To explore the impact of extracted shadow's accuracy on the results of building change detection, we uniformly sampled four subareas in the FZ and WH-B testing areas, each with a size of 512 × 512 pixels. We evaluated the accuracy of the extracted building shadows in terms of attribute and geometric measures (see Table VIII). Fig. 15 shows the distribution of TC error of the extracted shadows of these subareas and compares the corresponding false changes detected by the proposed SMCD-Net-m. We can see that, in the FZ testing area, for the subarea [e.g., in Fig. 15(d1)] having the extracted shadows with lower TC errors, building change detection results have fewer false changes, while obvious false changes were produced for subareas [e.g., Fig. 15(a1) and (b1)] where the shadows were inadequately extracted. A similar phenomenon can also be found in the WH-B testing area.

To further quantify the impact of shadow extraction accuracy on change detection results, we plotted the average value of F1 and GTC of the shadow extraction results from bitemporal images against the FP of change detection results for each subarea (see Fig. 16). It is clear that this figure shows that, with the increase in shadow extraction accuracy, the FP of the detected changes decreases accordingly. A more evident phenomenon can be found in the WH-B area because the bitemporal images in this area have a larger viewing angle than the FZ area. Moreover, more high-rise buildings, which were prone to cause false changes, were found in the WH-B area. Our results show a positive correlation between shadow extraction and change detection accuracies. This strongly

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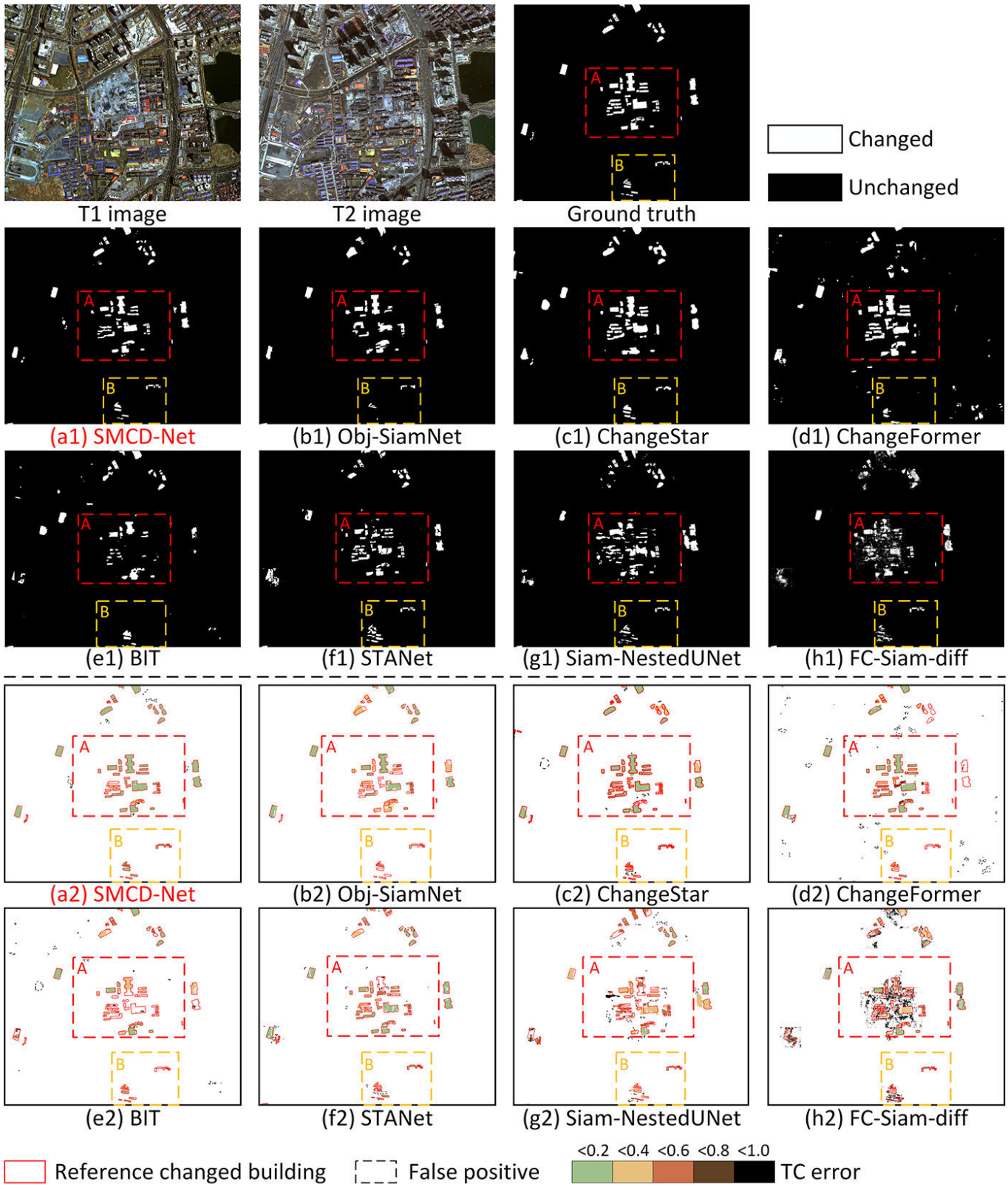


Fig. 10. (Top) Comparison of building change detection results between different methods in the WH-A testing area. (Bottom) Distribution of the TC error of different detection results. (a1) and (a2) SMCD-Net. (b1) and (b2) Obj-SiamNet. (c1) and (c2) ChangeStar. (d1) and (d2) ChangeFormer. (e1) and (e2) BIT. (f1) and (f2) STANet. (g1) and (g2) Siam-NestedUNet. (h1) and (h2) FC-Siam-diff.

proves the effectiveness of using the directional relationships between buildings and their shadows to alleviate false building change detections. Improving the shadow extraction accuracy is, thus, conducive to obtaining a more effective directional

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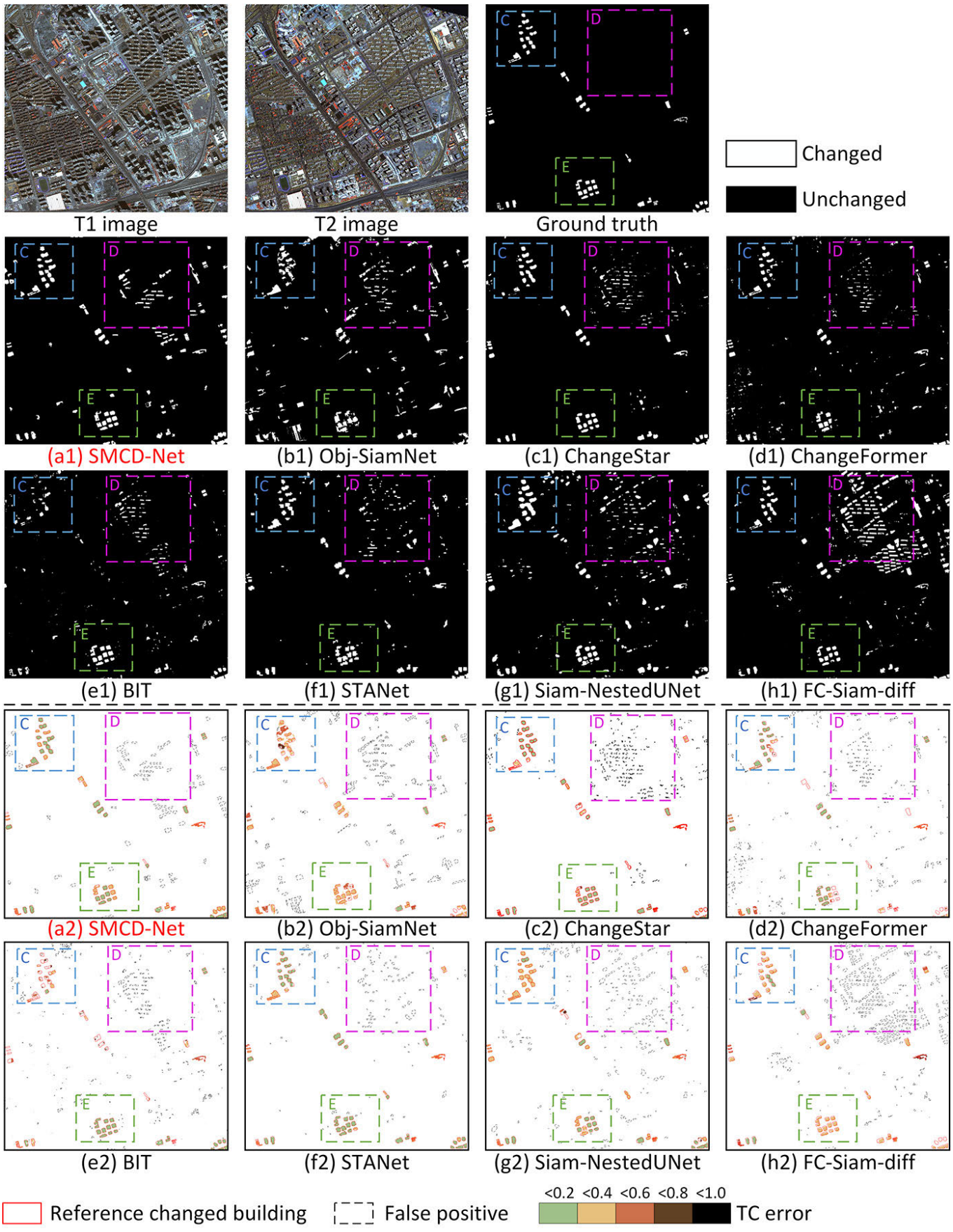


Fig. 11. (Top) Comparison of building change detection results between different methods in the WH-B testing area. (Bottom) Distribution of the TC error of different detection results. (a1) and (a2) SMCD-Net. (b1) and (b2) Obj-SiamNet. (c1) and (c2) ChangeStar. (d1) and (d2) ChangeFormer. (e1) and (e2) BIT. (f1) and (f2) STANet. (g1) and (g2) Siam-NestedUNet. (h1) and (h2) FC-Siam-diff.

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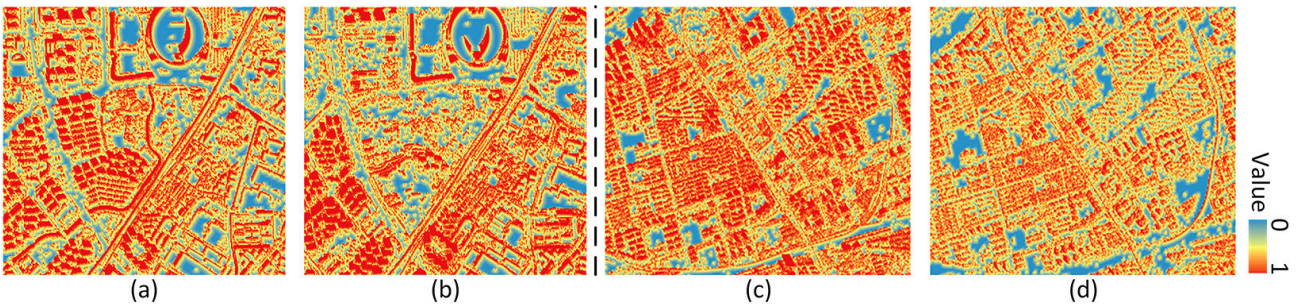


Fig. 12. Fuzzy landscape maps extracted from bitemporal VHR images in (a) and (b) FZ and (c) and (d) WH-B testing areas.

TABLE VI
EVALUATION OF THE CHANGES DETECTED BY DIFFERENT METHODS IN THE WH-A AND WH-B TESTING AREAS USING ATTRIBUTE INDICES AND GEOMETRIC ERRORS

Datasets	Methods	GOC	GUC	GTC	P	R	F1
WH-A	**SMCD-Net**	**0.057**	0.139	**0.113**	**0.894**	0.823	**0.857**
	Obj-SiamNet	0.082	0.209	0.179	0.864	0.762	0.809
	ChangeStar	0.194	**0.091**	0.160	0.789	**0.872**	0.828
	ChangeFormer	0.086	0.195	0.173	0.876	0.705	0.781
	BIT	0.102	0.239	0.211	0.858	0.538	0.662
	STANet	0.177	0.164	0.189	0.822	0.717	0.767
	Siam-NestedUNet	0.259	0.233	0.289	0.647	0.664	0.655
	FC_Siam_diff	0.141	0.327	0.265	0.538	0.529	0.533
WH-B	**SMCD-Net**	0.149	**0.018**	**0.109**	0.514	**0.881**	**0.649**
	Obj-SiamNet	0.158	0.077	0.132	0.283	0.754	0.412
	ChangeStar	0.116	0.063	0.112	**0.519**	0.806	0.631
	ChangeFormer	0.098	0.082	0.112	0.481	0.654	0.554
	BIT	**0.028**	0.225	0.163	0.345	0.312	0.328
	STANet	0.162	0.073	0.131	0.538	0.752	0.627
	Siam-NestedUNet	0.185	0.025	0.119	0.234	0.661	0.346
	FC-Siam-diff	0.165	0.036	0.128	0.537	0.807	0.517

TABLE VII
EVALUATION OF THE CHANGES DETECTED BY SMCD-NET-M AND SMCD-NET IN TERMS OF BOTH ATTRIBUTE INDICES AND GEOMETRIC ERRORS

Datasets	Methods	GOC	GUC	GTC	P	R	F1
FZ	SMCD-Net	**0.119**	0.236	0.213	0.774	0.737	0.754
	SMCD-Net-m	0.124	**0.167**	**0.171**	**0.782**	**0.780**	**0.781**
WH-B	SMCD-Net	0.149	**0.018**	0.109	0.514	0.881	0.649
	SMCD-Net-m	**0.112**	0.062	**0.105**	**0.620**	**0.895**	**0.751**

TABLE VIII
ACCURACY OF THE EXTRACTED BUILDING SHADOWS IN THE FZ AND WH-B AREAS

Datasets	Images	GOC	GUC	GTC	P	R	F1
FZ	T1	0.157	0.296	0.259	0.881	0.646	0.746
	T2	0.168	0.314	0.292	0.888	0.613	0.725
WH-B	T1	0.081	0.303	0.235	0.897	0.725	0.802
	T2	0.144	0.333	0.276	0.806	0.719	0.760

relationship, thereby improving the model's ability to distinguish false changes caused by different viewing angles.

V. DISCUSSION

This study investigates building change detection from bitemporal high-resolution satellite images. We developed a new SNN in a multitask learning framework (SMCD-Net) to improve detection accuracy, particularly in the geometric aspect. The MFE effectively integrates building boundary information to constrain the extraction of high-level semantic features from the geometric aspect. Our method performs better than many existing methods, i.e., Obj-SiamNet, ChangeStar, ChangeFormer, BIT, STANet, Siam-NestedUNet, and FC-Siam-diff. Moreover, to eliminate false changes caused by the difference in viewing angles of bitemporal images, we modeled directional relationships between buildings and the corresponding shadows into fuzzy landscapes by fuzzy sets and incorporated the directional information into SMCD-Net, leading to a network variant SMCD-Net-m. Our results confirmed that fuzzy landscapes are effective in characterizing directional relationships and suited for reducing false detection rates when using bitemporal images with obvious alignment errors.

High-resolution satellite images, characterized by fine spatial details, can improve the identification of small changes, while these images usually have a few numbers of spectral bands (e.g., four bands for most high-resolution satellites), leading to obvious detection noise when using pixel-based semantic segmentation methods. In this study, we incorporated a distance prediction task to constrain image feature extraction from the geometric aspect to guarantee the detected building changes have regularized boundaries. By doing so, the geometric accuracy of detected changes is evidently improved.

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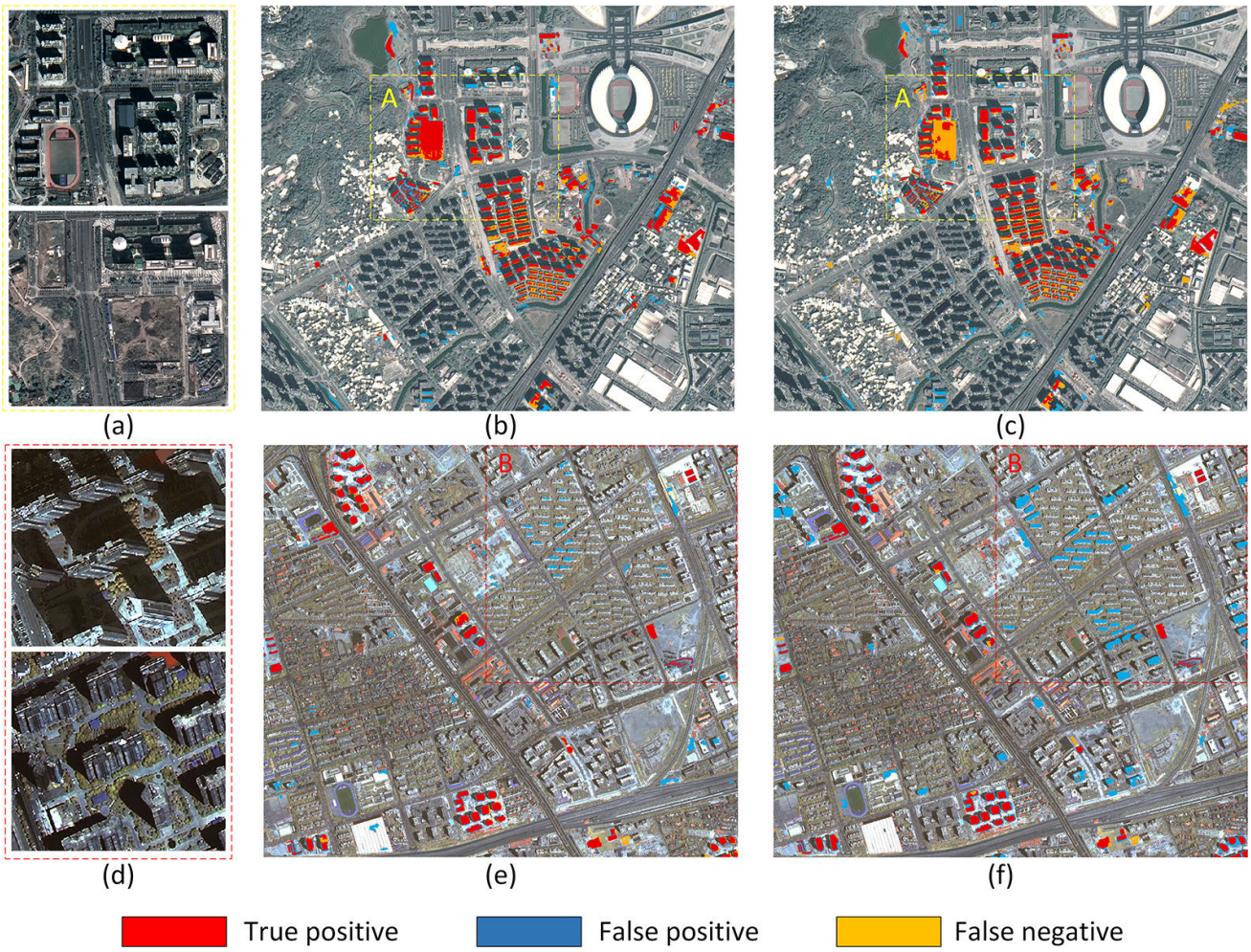


Fig. 13. Results of SMCD-Net-m and SMCD-Net in FZ and WH-B testing areas. (a) Examples of GF2 bitemporal images in the FZ testing area, and the corresponding detection results by (b) SMCD-Net-m and (c) SMCD-Net. (d) Examples of Pleiades bitemporal images on the WH-B testing area and the corresponding detection results by (e) SMCD-Net-m and (f) SMCD-Net.

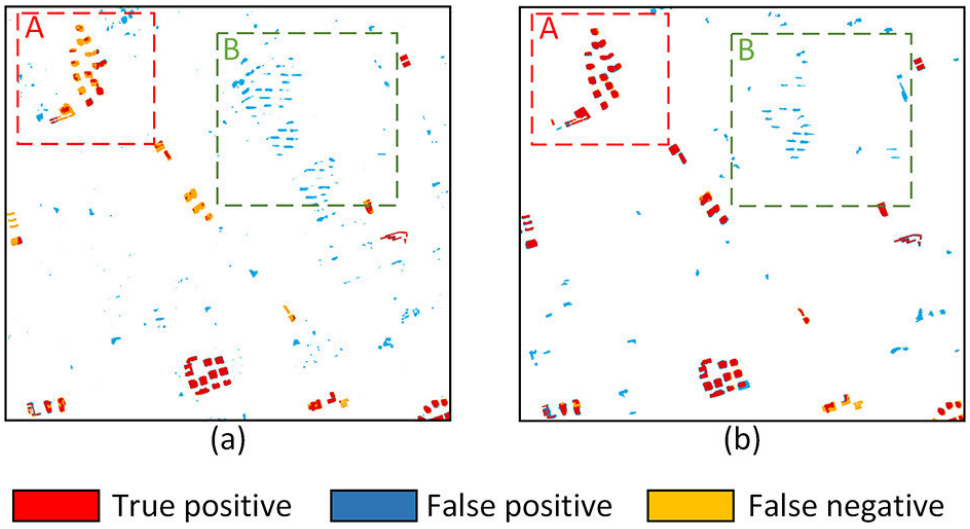


Fig. 14. Comparison of the results of different detection strategies in WH-B testing area. (a) Straightforward strategy treating fuzzy landscape map as an additional image channel. (b) Strategy used in SMCD-Net-m.

Compared with existing methods based on OBIA, our method has advantages in achieving automatic feature learning. On the other hand, the results of object-based methods are dependent on image segmentation, for which imperfect segmentation may increase detection uncertainty, particularly in complex urban environments.

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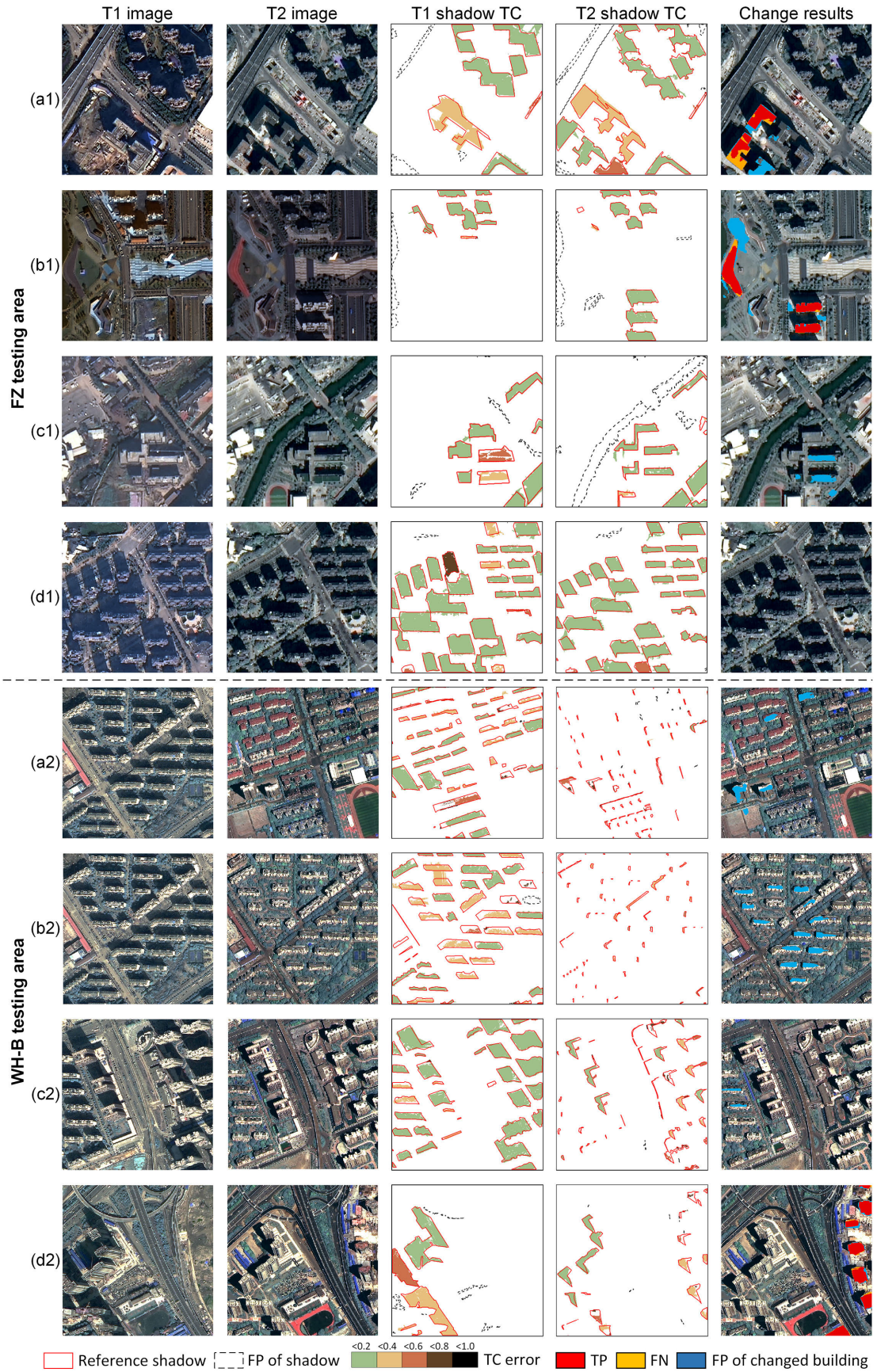


Fig. 15. (Top) TC distribution of the extracted shadows and the corresponding change detection results of four subareas (a1-d1) in the FZ testing area. (Bottom) TC distribution of the extracted shadows and the corresponding change detection results of four subareas (a2-d2) in the WH-B testing area. The reference shadow data were obtained by manually digitalizing the target VHR images.

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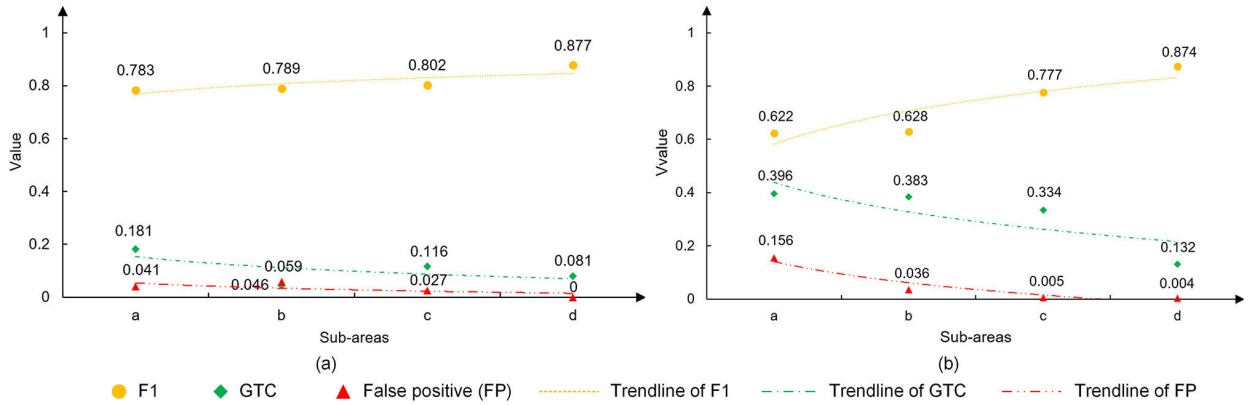


Fig. 16. (a) and (b) Correlation analysis between the accuracy of shadow extraction and change detection in FZ and WH-B testing areas, respectively. Trendlines were fit via logarithmic functions.

TABLE IX

ACCURACY OF THE EXTRACTED BUILDING SHADOWS BY EFFICIENTNET IN THE FZ AND WH-B AREAS

Datasets	Images	GOC	GUC	GTC	P	R	F1
FZ	T1	0.077	0.150	0.133	0.830	0.842	0.746
	T2	0.060	0.171	0.138	0.911	0.817	0.725
WH-B	T1	0.074	0.122	0.115	0.913	0.861	0.802
	T2	0.105	0.205	0.165	0.857	0.695	0.767

TABLE X

EVALUATION OF THE CHANGES DETECTED BY SMCD-NET-M USING SHADOW INFORMATION EXTRACTED BY EFFICIENTNET IN TERMS OF BOTH ATTRIBUTE INDICES AND GEOMETRIC ERRORS

Datasets	Methods	GOC	GUC	GTC	P	R	F1
FZ	SMCD-Net-m	0.121	0.127	0.143	0.764	0.820	0.792
WH-B	SMCD-Net-m	0.113	0.041	0.95	0.675	0.938	0.785

A. Effectiveness of Directional Relationships for Reducing False Changes Caused by Different Viewing Angles

A major issue exists in building change detection using bitemporal VHR images, i.e., the difference of their viewing angles brings evident false detected changes due to the spatial displacement of tall objects, such as buildings. To deal with this issue, we first modeled the directional relationships between buildings and the corresponding shadows by fuzzy landscapes. Then, we designed a lightweight network to extract high-level directional features based on fuzzy landscape maps to determine to what extent the detected building changes belong to false changes. The lightweight network is also embedded into the proposed change detection network, and the fuzzy landscape maps are used as multimodal data. By doing so, false changes are evidently reduced, further improving building change detection. Differing from existing methods using DSM to deal with the above issue, our method is more applicable to change detection only using VHR images.

B. Better Shadow Extraction Contributes to the Improvement of Change Detection

Our results identified a positive correlation between shadow extraction and change detection accuracies (see Fig. 16). In this study, we used an unsupervised method based on the Otsu threshold to extract shadow areas from VHR multitemporal images [51], [52]. This method has been popularly used for many applications [52], [61], [62] and achieved satisfactory results in our study. Nevertheless, we expect an improvement in change detection results when using a more advanced shadow extraction method based on, for example, deep learning. To verify this expectation, we used EfficientNet [63], which is a powerful semantic segmentation model, to extract shadow areas. Here, we collected training samples from the target VHR images by manual interpretation and trained the model for 100 epochs with a learning rate of 10^{-3} and a batch size of 16. Table IX lists the accuracy of shadow extraction results by EfficientNet in the FZ and WH-B testing areas, and Table X evaluates the changes detected by SMCD-Net-m using the extracted shadow information. Comparing with the shadow extraction results by the Otsu method (see Table VIII) and the corresponding change detection results (see Table VII), we can see that EfficientNet extracted shadows with higher accuracy than that of the Otsu method, leading to an accuracy improvement in the final change detection. Fig. 17 compares shadow extraction and change detection results between the Otsu and EfficientNet methods in the two testing areas. This figure shows that the false changes caused by different viewing angles were further reduced when using more accurate shadow information produced by EfficientNet.

TABLE XI

COMPARISON OF THE NUMBER OF PARAMETERS (PARAMS) AND FLOPS BETWEEN DIFFERENT CHANGE DETECTION MODELS

Methods	Backbones	Params (M)	FLOPs (G)
SMCD-Net-m	UNet	7.90	9.53
Obj-SiamNet	ResNet34	21.80	7.34
ChangeStar	ResNet50	33.95	11.75
ChangeFormer	MiT-b1	24.30	16.84
BIT [60]	ResNet18	3.55	17.40
STANet	ResNet18	16.92	26.03
Siam-NestedUNet [58]	UNet++	12.03	109.69
FC-Siam-diff	UNet	1.35	9.45

C. Analysis of Model Complexity

We compared the number of parameters and floating point operations (FLOPs) between different change detection

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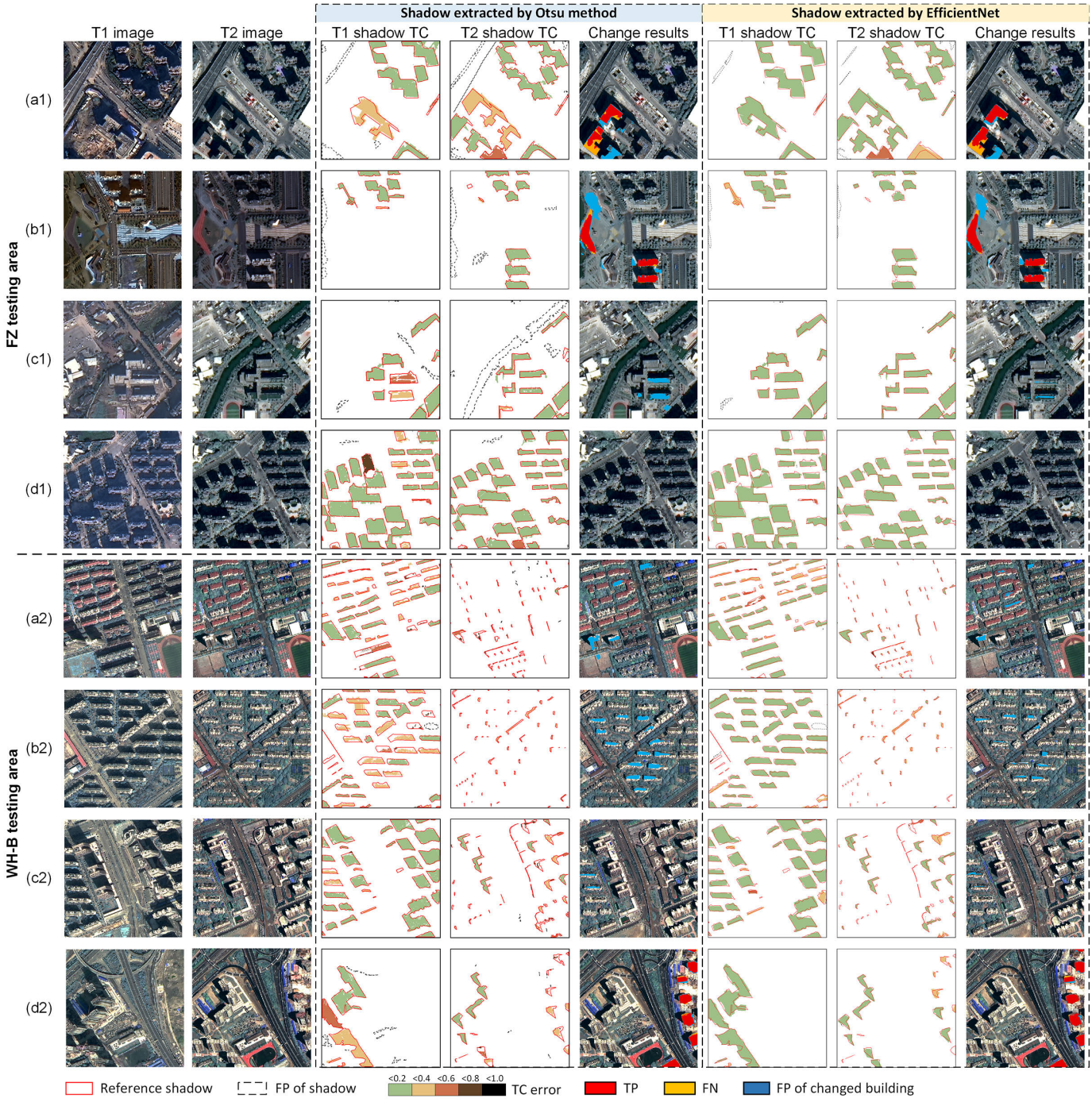


Fig. 17. Comparison of shadow extraction and change detection results between the Otsu and EfficientNet methods for different subareas in (top) FZ testing area and [bottom WH-B testing area, corresponding to (a1-d1) and (a2-d2)]. The reference shadow data were obtained by manually digitalizing the target VHR images.

models (see Table XI). We were also aware that many previous studies (e.g., [9], [31], and [59]) did not provide adequate information for model's complexity. To make a fair comparison, we used the *thop* package (available at https://github.com/Lyken17/pytorch-OpCounter) in PyTorch to estimate the number of parameters and FLOPs for each model with the same-size input image of 256 × 256 pixels. Among many, the proposed SMCD-Net-m has relatively fewer numbers of parameters and FLOPs, indicating its efficiency for implementation. This table also shows that the models based on UNet backbones are more lightweight than those based on ResNet and transformer.

D. Effectiveness of Transferring Samples by CycleGAN

To enrich training samples, we used CycleGAN to transfer samples from a source dataset to the target dataset. We transferred FZ GF-2 samples with a 0.8 m spatial resolution to detect building changes from 0.5 m Pleiades images in the WH area. Using the transferred samples, the accuracy of detection in the WH area was evidently improved (see Table V). Our

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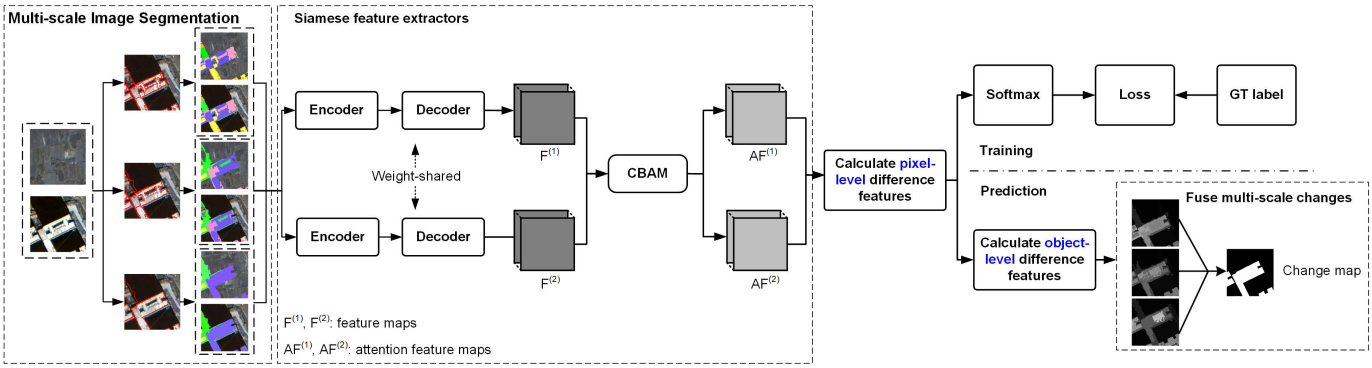


Fig. A1. Architecture of the Obj-SiamNet.

results confirmed the effectiveness of the transferring method. In this study, the spatial resolution of a GF-2 image is similar to a Pleiades image. The effectiveness of applying CycleGAN to transfer samples from a source dataset with a larger resolution different, however, is unexplored. This leads to our future study.

VI. Conclusion

This study presents an SNN based on multitask learning architecture (i.e., SMCD-Net) to detect building changes from bitemporal high-resolution remote sensing images. We constructed a SMFE to integrate boundary information for regularizing detected building changes, leading to accuracy improvement in both attribute and geometric aspects. To address the problem of false changes caused by the spatial displacement of tall objects in bitemporal images with different viewing angles, we modeled directional relationships between buildings and their shadows by fuzzy landscape maps, used as multimodal data, corresponding to SMCD-Net-m. Experiments were conducted on a public CD_Data_GZ dataset, a GF-2 VHR dataset in FZ, and a Pleiades dataset in WH. We concluded the following.

1) Compared with the existing models (i.e., Obj-SiamNet, ChangeStar, ChangeFormer, BIT, STANet, SiamNestedUNet, and FC-Siam-diff), SMCD-Net achieved the best detection results on these three datasets, with an F1-score up to 0.917 on the public GZ dataset. It also produced the lowest global TC errors on all the datasets, implying the effectiveness of the proposed MFE for improving the geometric accuracy of detected changes.
2) By using directional information, the detection results by SMCD-Net-m were evidently improved on the WH dataset, reducing the false changes caused by different viewing angles of bitemporal images effectively. The improvement was more evident for bitemporal VHR images with a larger difference in viewing angles.
3) The accuracy of shadow extraction affects the subsequent change detection when incorporating directional information, showing a positive correlation between the accuracy of shadow extraction and change detection.
4) CycleGAN can be effectively used to transfer source samples to the target domain to enrich training samples. In the WH study area, the accuracy was improved by 30.6% on average compared with those without sample transferring.

Appendix

See Fig. A1.

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